Global Vision-Language Feature Interaction Enhanced by Object-Context Association for Remote Sensing Visual Grounding

Jun Xie, Bing Zhang, Zhengchao Chen, Xuan Yang, *Member, IEEE*, Yongqing Bai, Zhaoming Wu, and Yue Xu

***Abstract*—Remote sensing visual grounding (RSVG) aims to accurately localize specific targets in remote sensing (RS) images based on natural language descriptions. However, existing RSVG datasets often contain overly simplistic textual descriptions, exhibit imbalanced object size distributions, and lack semantic connections between targets and the surrounding contexts. Moreover, current approaches rely on global visual features for region-level localization, while the interaction between visual and textual modalities remains limited. To address these challenges, we propose improvements from both the dataset and algorithm perspectives. First, we explore the semantic correlation between RS objects and their scene context. Based on high-resolution Gaofen satellite imagery, we expand several typical object categories and construct richer textual descriptions that reflect object-background associations. By refining and extending the existing DIOR-RSVG dataset, we build a new dataset named DGF-RSVG. Second, to enhance the semantic alignment between global visual features and textual features, we propose a novel global vision-language multimodal feature interaction enhancement (GME) module. In parallel, we design a local attention enhancement (LAE) module to facilitate fine-grained interaction between object-related textual features and regional visual proposals. These two modules form the foundation of our newly developed detection framework: the global–local attention enhanced detector (GLAED). Extensive experiments show that GLAED achieves state-of-the-art performance on the DGF-RSVG dataset, outperforming the closest competitor by 5.01% in Pr@0.5. It also achieves highly competitive results on the DIOR-RSVG dataset, demonstrating the effectiveness of both our proposed dataset and model framework.**

***Index Terms*—Global–local attention, multimodal interaction, object-context association, remote sensing visual grounding (RSVG).**

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I. Introduction

In RECENT years, the launch of numerous remote sensing (RS) satellites has provided massive volumes of high-resolution optical imagery for Earth observation. Traditional manual interpretation, which depends on expert knowledge to extract target information from large-scale RS data, is highly inefficient. The emergence of deep neural networks, inspired by the structure of biological neurons [1], along with advancements in hardware computing capabilities [2], has greatly enhanced the automation level of image information extraction. Research on object detection in natural images has aligned well with the analytical demands of RS data. By adapting these techniques to the specific characteristics of RS imagery, researchers have significantly advanced the development of RS object detection (RSOD) methods. With the growing maturity of natural language processing [3], textual features have become a valuable and stable source of semantic information, which helps to address the limitations of visual representations [4], [5], [6]. As a result, RSOD has evolved from relying solely on visual features to incorporating multimodal information by combining image and textual features [7], [8], [9]. This transition has further improved the intelligence and robustness of RS interpretation. The rapid development of RSOD algorithms, characterized by efficient image processing and precise object localization, provides strong support for a wide range of applications, including land planning, ecological monitoring, and disaster response [10], [11], [12].

RS visual grounding (RSVG) is an emerging research direction in the field of RSOD. Sun et al. [13] first introduced the visual grounding task into RS in 2022. Considering that RS images often contain large-scale scenes and rich geospatial information about ground objects, they employed textual

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language to describe RS targets and collected a new visual grounding dataset named RSVG. They further proposed a model named GeoVG, which consists of a language encoder for processing geospatial relations, an image encoder with adaptive region attention, and a fusion module for multimodal interaction. However, this work mainly focuses on quantifying the spatial relationships between objects, while in practical scenarios, it is often difficult to provide precise numerical distance descriptions. Subsequently, Zhan et al. [14] extended the DIOR [15] dataset by annotating each object with textual expressions, thereby constructing DIOR-RSVG, which has become one of the most widely used datasets for RSVG tasks. They also introduced a novel Transformer-based [16] multigranularity visual language fusion (MGVLF) module, which advanced the research in this area. Nevertheless, the textual descriptions in DIOR-RSVG are relatively simple and primarily focus on the relative position of objects within the image, while overlooking the semantic relevance between targets and their surrounding background. Moreover, there is still room for improvement in the cross-modal feature fusion capabilities of MGVLF.

RSVG demonstrates its notable advancement by enabling object localization in RS imagery through natural language queries. Unlike conventional RSOD methods that rely solely on unimodal visual features, RSVG adopts a distinct task paradigm in two key aspects. First, it introduces and leverages cross-modal textual features. Traditional visual object detection models are typically trained on unimodal image data, learning to localize objects based solely on visual cues. In contrast, RSVG builds upon standard RSOD annotations by incorporating textual descriptions of objects within the image. It augments the network architecture with a language encoder to extract textual features and facilitate their interaction with visual features. This enables the model to perform target localization under the guidance of language, thereby making the detection process more intelligent. However, this cross-modal design inevitably increases model complexity and imposes greater demands on the network's ability to align heterogeneous features effectively. Second, RSVG supports a more open-ended form of detection. Conventional detectors are trained on fixed category labels using bounding-box annotations and can only identify objects belonging to a predefined set of classes. In contrast, RSVG allows a single object to be described in various ways, such as by its size, color, or relative position, using flexible natural language expressions. By accepting these diverse descriptions as query prompts, RSVG significantly enhances the openness and flexibility of object detection in RS scenarios.

Research on the RSVG task has led to promising developments in methods and datasets. However, several challenges remain to be addressed.

First, the quality of textual descriptions in existing RSVG datasets remains insufficient.

1) Current descriptions are overly simplistic, often following a fixed pattern such as *"A/An + color/shape/relative position + object name."* For example, in the DIOR-RSVG dataset, a large number of expressions rely on relative position cues. In real-world scenarios, such positional relationships are often unstable and vary with context. Since textual descriptions serve as an alternative labeling mechanism for visual objects, the core objective of RSVG is to align these descriptions with specific regions in the image. Overly simple expressions fail to capture the distinct characteristics of different instances, reducing the model's ability to generalize.
2) Existing datasets largely ignore the contextual relationship between targets and their background. While traditional visual detection models treat background regions as negative samples, RSVG datasets often omit background information in the descriptions. In practice, RS targets frequently exhibit strong spatial associations with specific background environments. For example, ships are typically docked at ports or travel through waterways, ground track fields are often surrounded by residential areas, and wastewater treatment plants are commonly located near factories or rivers. Similar objects may also appear in various geographical scenes, such as wind turbines in deserts, hills, or offshore areas, and bridges in valleys, across rivers, or within urban and rural regions. These spatial correlations reflect the inherent link between RS targets and their surrounding context, and incorporating this information into textual descriptions provides richer semantic guidance.

Second, the current approaches to multimodal feature interaction remain limited. In RSVG, the textual query typically refers to a local region, yet the language features are often fused with global visual features. Achieving effective alignment between these two modalities at the image level is key to precise localization. Existing models commonly adopt dual-branch architectures to process visual and textual inputs independently, followed by straightforward fusion strategies such as feature addition or simple attention mechanisms. However, these methods often overlook the hierarchical, multiscale nature of visual features resulting from downsampling operations, and they fail to capture the fine-grained dynamics within the language features. As a result, localization performance is significantly constrained. To overcome these challenges, we conduct comprehensive research from both dataset construction and model design perspectives. The main contributions of our work are summarized as follows.

1) To address the limitations of existing RSVG datasets, including overly simplistic textual descriptions, extreme variations in target size, and insufficient consideration of the contextual relationships between targets and their surrounding environments, we construct a new RSVG dataset named DGF-RSVG. Specifically, we enrich the diversity of target categories and scenes using Gaofen satellite imagery, enhance textual annotations by incorporating semantic associations between targets and their geographic contexts, and refine the DIOR-RSVG dataset through targeted sample selection.
2) To enhance the capability of visual–language feature interaction in RSVG, we propose a novel RSVG model, global–local attention enhanced detector (GLAED), which incorporates two key modules: a global

vision-language multimodal feature interaction enhancement (GME) module to facilitate image-level alignment between visual and textual features, and a local attention enhancement (LAE) module to improve the interaction between region-level visual features and text features.

3) We conducted experiments on the DGF-RSVG and DIOR-RSVG datasets, and the results show that the proposed GLAED outperforms the state-of-the-art RSVG methods. On the DGF-RSVG dataset, GLAED achieves a 5.01% improvement in Pr@0.5 over the closest competitor. It also achieves highly competitive results on the DIOR-RSVG dataset.

II. Related Works

In this section, we present a comprehensive review of visual grounding research in both natural and RS images. Section II-A discusses representative datasets and key methodologies in natural images, while Section II-B outlines recent advances in RSVG.

A. Visual Grounding in Natural Images

Visual grounding tasks require the joint participation of images, referring expressions, and corresponding bounding-box annotations. The introduction of the ReferItGame [17] dataset sparked research interest in the understanding of referring expressions, bringing VG into practical scenarios. Subsequently, the RefCOCO/+/g [18], [19] datasets, which provide object-level referring expressions based on MSCOCO [20], offered more comprehensive and reliable benchmarks for VG tasks. With the growth of large-scale data and the success of deep learning, convolutional neural networks (CNNs) [21] and long short-term memory (LSTM) [22] networks have driven the integration of computer vision and natural language processing. By extending object detection frameworks with language processing branches, VG models have enabled joint visual-linguistic reasoning and have developed rapidly. According to the differences in architectural design, existing VG methods can be categorized into three main groups.

1) Two-Stage Approaches: As illustrated in Fig. 1(a), these methods first use pretrained object detectors to generate region proposals and then match them with textual descriptions in a second stage. The core objective of these approaches lies in accurately computing the similarity between candidate regions and textual descriptions. Hu et al. [23] proposed a novel spatial context recurrent convnet, which incorporates spatial information among objects and global scene context to estimate the probability of each candidate region under the given query. Yu et al. [24] developed the modular attention network, which dynamically models object appearance, spatial location, and relationships with surrounding objects through language-guided and visual attention mechanisms, thereby enhancing the understanding of visual regions. In addition, tree networks [25], [26] and graph attention networks [27], [28] have been introduced to further improve region-text similarity modeling. However, these methods heavily depend on the quality of the generated region proposals and often incur high computational costs.

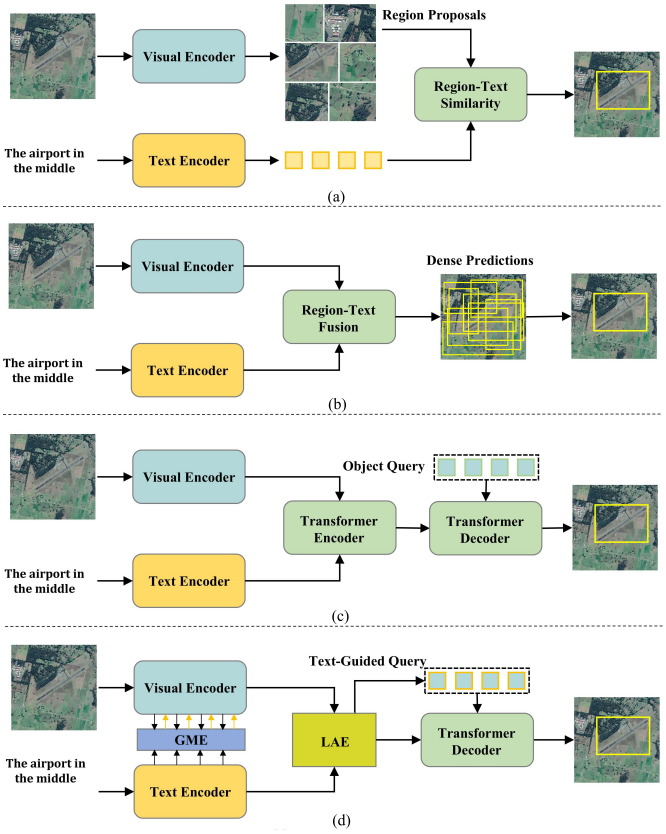


Fig. 1. RSVG frameworks with different processing strategies. (a) Two-stage pipeline relying on pretrained weights to calculate the similarity of image and text. (b) One-stage pipeline with dense prediction after fusion of visual and language features. (c) Transformer-based pipeline through object queries. (d) Our method focusing on the interaction between image-level and local visual features.

2) One-Stage Approach: As illustrated in Fig. 1(b), one-stage approach aim to improve computational efficiency by discarding the region proposal stage and directly performing dense predictions on fused visual-linguistic feature maps. Chen et al. [29] introduced a guided attention mechanism that encourages the model to focus on the central region of the referent. Based on YOLOv3 [30] and BiLSTM [31], they developed an end-to-end single-stage grounding network. Building on this idea, Yang et al. [32] integrated BERT-based [33] textual embeddings into the YOLOv3 detector and enhanced them with spatial features, proposing a simple and efficient model called FAOA. Recognizing that encoding a query as a single sentence embedding may lose fine-grained details, Yang et al. [34] further proposed ReSC, a representative model that performs iterative reasoning between the image and recursive subqueries to reduce matching ambiguity. Despite improved efficiency, one-stage methods often struggle with capturing fine-grained object details due to the dense prediction nature.

3) Transformer-Based Approaches: As shown in Fig. 1(c), the success of self-attention mechanisms in Transformer architectures has shifted the focus of VG research from CNNs and LSTMs to more advanced Transformer-based models. TransVG [35] pioneered the use of the Transformer in VG, reformulating it as a coordinate regression task and achieving

superior accuracy compared with two-stage and one-stage methods. Deng et al. [36] proposed TransVG++, addressing optimization limitations in TransVG by designing a language-conditioned vision Transformer. VLTVG [37] introduced a language-guided feature encoder and multistage cross-modal decoders for target retrieval. Further improving cross-modal alignment, Ye et al. [38] proposed QRNet with a novel query-aware dynamic attention and multiscale fusion mechanism to refine visual features for grounding tasks, achieving compelling results.

B. Visual Grounding in RS Imagery

Sun et al. [13] constructed the first RSVG dataset by collecting 7699 satellite and aerial images paired with corresponding textual descriptions, introducing the VG task into the RS domain. They proposed GeoVG, which incorporates geospatial relationships into the language encoder; however, its reliance on geospatial distances limits practical applicability. Considering the challenges of scale variation and complex backgrounds in RS images, Zhan et al. [14] designed a novel Transformer-based MGVLF module. To address the lack of RSVG datasets, they also developed a large-scale benchmark named DIOR-RSVG based on the optical RS dataset DIOR. Recent studies have increasingly focused on RSVG. Wang et al. [39] proposed a multistage synergistic aggregation module (MSAM) to overcome the limitations of simplistic vision-language fusion strategies. Inspired by DETR, Lan et al. [40] introduced LQVG, a Transformer-based framework that uses sentence-level textual features as queries. This method currently achieves state-of-the-art performance on the DIOR-RSVG dataset. To address the lack of interaction between visual and textual features during visual encoding, Li et al. [41] proposed a language-guided progressive visual attention (LPVA) mechanism and constructed a new dataset, OPT-RSVG, for understanding complex scenes. In addition, several other methods, such as QAMFN [42], RINet [43], and FQRNet [43], have contributed to the field. Most recently, Hu et al. [44] proposed an open-set RSVG approach to address the challenge of multiobject prediction in RS images. They introduced a multiscale image-to-text fusion module (MSITFM) and a text confidence matching (TCM) mechanism to mitigate background interference. However, existing methods often overlook the semantic correlations between foreground objects and background regions, treating backgrounds merely as distractors, which limits the full potential of RSVG.

III. Methods

In this section, we present the proposed GLAED framework for RSVG. The overall architecture is introduced in Section III-A. The GME module is described in Section III-B. Section III-C details the LAE module. The multimodal Transformer decoder is presented in Section III-D.

A. Overview

The proposed GLAED is designed to achieve accurate object localization in RS images by effectively integrating visual content with corresponding textual descriptions. As shown in the general structure diagram of Fig. 1(d), unlike previous methods that simply perform addition or basic interaction between visual and textual features, GLAED first emphasizes the interaction between image-level visual features and textual semantics to establish global associations. Based on this design principle, the GME module is introduced. Subsequently, the model shifts its attention to local regions for object-level grounding, which is achieved through the LAE module.

As illustrated in Fig. 2, the detailed architecture of GLAED consists of a dual-stream architecture. The top-left branch represents the text processing stream. Text descriptions containing object-related scene information are first tokenized into word embeddings. These embeddings are then fed into a text backbone BERT to extract textual features $F_T \in \mathbb{R}^{l\times d}$, where l and d denote the number of tokens and linguistic feature dimension, respectively. The bottom-left branch shows the visual processing stream. Given an RS image $I \in \mathbb{R}^{H\times W\times 3}$, the visual backbone ResNet [45] is used to extract visual features F_V. Shallow-level features interact with textual features through the GME module in a progressive manner to generate deeper features. As a result, multiscale feature maps $F_{V/i} \in \mathbb{R}^{(H/i)\times(W/i)\times C}$ are obtained with strides $i \in \{8, 16, 32, 64\}$. The multiscale features with unified dimension C are then fused with the textual features F_T via multiple attention modules to strengthen cross-modal interaction, yielding enhanced features $F'_{V/i}$ and F'_T. Based on these, the LAE module generates candidate regions from the visual stream and filters them through interaction with the textual features for category alignment. Subsequently, the selected candidate features, along with the multiscale visual features and the textual features, are fed into a multimodal Transformer decoder for further cross-modal fusion. Finally, the decoder outputs are passed through classification and regression branches to predict the object category and bounding box. The loss is computed with respect to ground truth for training through backpropagation, enabling accurate object classification and localization.

B. GME Module

RS imagery provides a spatial representation of ground surface compositions, encompassing both object appearance and its associated contextual scene. Automatically distinguishing objects within complex distributional scenes based solely on visual features remains challenging. To mitigate this issue, textual descriptions are introduced to supplement objects with diverse contextual labels, enabling stronger coupling between objects and their corresponding scenes, while suppressing irrelevant visual distractions. However, RS images and text descriptions inherently belong to different modalities. Establishing effective interactions between these heterogeneous data types is essential for aligning them semantically. In this work, geographic scene knowledge related to object distributions is introduced into the network in the form of textual inputs. These descriptions interact with multiscale visual features through attention mechanisms to build high-granularity cross-modal correlations.

Given a text description $T = \{w_1, w_2, \ldots, w_l\}$ associated with an RS image, where $0 < l \leq 40$, the sequence is

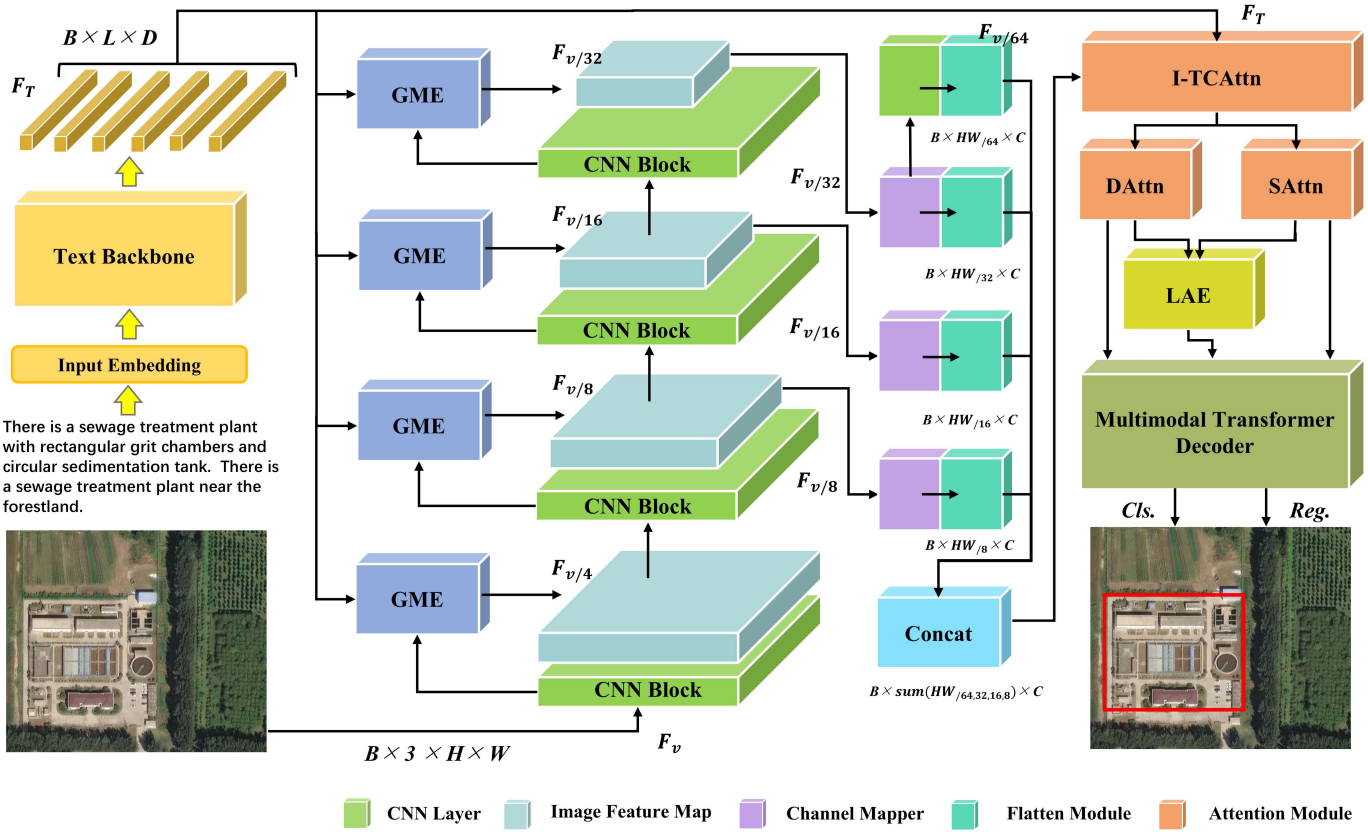


Fig. 2. Overall structure of our proposed GLAED. It comprises a text backbone, ResNet-based visual backbone, GME module, dimension transformation module, LAE module, and multimodal Transformer decoder. The text and visual backbones extract features from the input description and RS image, respectively, and encode them as vectors. The GME module performs image-level interaction between language features and multiscale visual features to generate enhanced visual representations. These are integrated by the dimension transformation module through linear projection and concatenation. The LAE module then refines region-specific visual features through local-level multimodal interaction. Finally, all features are deeply fused in the multimodal Transformer decoder to predict the coordinates of the target.

first tokenized and augmented with special tokens to form $T' = \{[\text{CLS}], w_1, w_2, \ldots, w_l, [\text{SEP}]\}$. The tokenized sequence is truncated if it exceeds the maximum length of 40. Each token is then converted into an embedding vector, yielding $T_{\text{Emb}} \in \mathbb{R}^{l \times d_t}$, where d_t denotes the embedding dimension. These embeddings are fed into a BERT-based text backbone to extract semantic features $F_T \in \mathbb{R}^{l \times d}$, with the output feature dimension d set to 768. For the corresponding RS image $I \in \mathbb{R}^{H \times W \times 3}$, a CNN backbone is employed to extract hierarchical visual representations. After five successive CNN blocks, multiscale visual features are generated at different strides.

The detailed processing of the GME module is illustrated in Fig. 3. Taking the first GME block as an example, the input text feature is denoted as $F_T \in \mathbb{R}^{B \times l \times d}$, where B is the batch size. It is first projected through a linear layer to obtain the transformed feature $F_{T'} \in \mathbb{R}^{B \times l \times d'}$. The process is expressed as follows:

$$F_{T'} = \mathcal{F}_{\text{Linear}}(F_T). \tag{1}$$

Given the visual feature map from the previous CNN block, denoted as $F_{v_i} \in \mathbb{R}^{B \times (H/4) \times (W/4) \times C}$, it is first flattened to form $F_{v_1} \in \mathbb{R}^{B \times (H/4)(W/4) \times C}$ to facilitate interaction with the text features. To enhance the regional perception capability, deformable attention [46] is applied to F_{v_1}, enabling adaptive

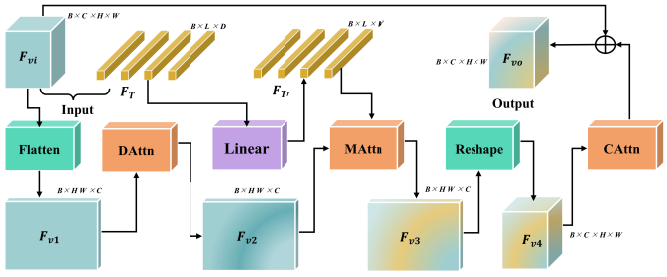


Fig. 3. Illustration of the proposed GME module. Attention computation with language features is performed on input visual features in a shared space, followed by updating and multilevel interactions to enhance visual representation.

spatial sampling that captures the diverse scales and shapes of objects in RS imagery. It allows the model to focus on relevant regions, improving alignment between visual and textual features. The resulting output $F_{v_2} \in \mathbb{R}^{B \times (H/4)(W/4) \times C}$ encodes enhanced visual representations with improved regional awareness. The processing can be formulated as follows:

$$F_{v_1} = \mathcal{F}_{\text{Flatten}}(F_{v_i}), \quad F_{v_2} = \mathcal{F}_{\text{DAttn}}(F_{v_1}). \tag{2}$$

A multimodal attention mechanism is then adopted to fuse the text and visual features. In this process, the visual features

F_{v_2} act as queries, while the transformed text features $F_{T'}$ serve as keys and values. Prior to attention computation, the channel dimensions C of F_{v2} and the dimension d' of $F_{T'}$ are projected to a common embedding dimension to ensure compatibility. The multimodal attention output is computed as follows:

$$F_{v3} = \mathcal{F}_{\text{MAttn}}(F_{v2}, F_{T'}) = \text{softmax}\left(\frac{F_{v2}F_{T'}^{\top}}{\sqrt{d_k}}\right)F_{T'} \quad (3)$$

where d_k denotes the dimension of the key vectors and is used to scale the dot product for numerical stability, following the standard attention mechanism.

The output F_{v_3} is then reshaped back to match the original spatial size of the feature map

$$F_{v_4} = \mathcal{F}_{\text{Reshape}}\left(F_{v_3}\right). \quad (4)$$

To further enhance the global expressiveness of visual features, channel attention [47] is applied to F_{v_4}. A residual connection is also added to preserve the original visual information, yielding the final output of the GME block

$$F_{v_o} = F_{v_i} + \mathcal{F}_{\text{CAttn}}\left(F_{v_4}\right). \quad (5)$$

The output F_{v_o} is then passed into the next CNN block to extract higher level visual features. This GME process is repeated at multiple scales to enable global vision-language interaction at different resolutions, ultimately generating multiscale visual feature maps with strides of $i \in \{8, 16, 32\}$, denoted as $F_{V/i} \in \mathbb{R}^{(H/i)\times(W/i)\times C}$.

C. LAE Module

In the proposed framework, the GME module facilitates global-level interaction between visual and textual features extracted from RS images, enabling effective alignment between descriptive language and multiscale global visual representations. However, the visual grounding task inherently demands precise localization of specific objects or regions, which calls for fine-grained analysis of local visual areas. To this end, we design an LAE module that explicitly strengthens the network's sensitivity to local visual details for accurate grounding. The LAE module takes as input the text features $F_T \in \mathbb{R}^{B\times l\times d}$ obtained from the text backbone and the multiscale visual features $F_{V/i} \in \mathbb{R}^{B\times(H/i)\times(W/i)\times C}$ with strides $i \in \{8, 16, 32, 64\}$. Here, B denotes the batch size, and the stride-64 feature map is directly downsampled from the stride-32 output of the GME module.

Before being fed into the LAE module, the multiscale visual features undergo a series of preprocessing steps to ensure compatibility with textual representations. Specifically, a channel mapper is applied to unify the channel dimension across all scales to a common size C. Then, each feature map is reshaped using a flattening operation from $[B, H, W, C]$ to $[B, HW, C]$, enabling efficient sequential processing. The processed features from all four scales are then concatenated along the spatial dimension to form the unified visual input to the LAE module, denoted as F_{vl}

$$F_{vl} = \mathcal{F}_{\text{Concat}}\left(\mathcal{F}_{\text{Flatten}}\left(\mathcal{F}_{\text{ChannelMapper}}\left(F_{V/8}, F_{V/16}, F_{V/32}\right)\right), \mathcal{F}_{\text{Flatten}}\left(\mathcal{F}_{\text{Conv}}\left(F_{V/64}\right)\right)\right). \quad (6)$$

This processing yields a compact and unified representation $F_{vl} \in \mathbb{R}^{B\times N\times C}$, where $N = \sum_{i\in\{8,16,32,64\}}(H/i)(W/i)$, effectively aggregating multiscale spatial information while aligning feature dimensions for subsequent interaction with textual guidance.

The detailed workflow of the LAE module is illustrated in Fig. 4. First, the visual features F_{vl} and textual features F_T are fed into the cross-attention module I-TCAttn, enabling interactive updates between the two modalities. Specifically, F_{vl} is used as the query to attend to F_T, resulting in the updated textual feature F_{T1}. Then, F_{T1} serves as the query to attend back to F_{vl}, producing the updated visual feature F_{vt1} and refined textual feature F_{T1}, The computation is expressed as follows:

$$F_{T1} = \mathcal{F}_{\text{MAttn}}(F_{vl}, F_T), \quad F_{vt1} = \mathcal{F}_{\text{MAttn}}(F_{vl}, F_{T1}). \quad (7)$$

Following the cross-modal refinement, the updated features are further processed by modality-specific attention mechanisms. The visual features F_{vt1} are enhanced using deformable attention to better capture spatial variability, while the textual features F_{T1} are refined through a self-attention mechanism to preserve and enhance their internal semantic structure. This refinement facilitates the interpretation of descriptions involving spatial relations or fine-grained attributes, thereby maintaining the consistency and contextual coherence of textual guidance throughout the grounding process. The operation is formulated as follows:

$$F_{vt2} = \mathcal{F}_{\text{DAttn}}(F_{vt1}), \quad F_{T2} = \mathcal{F}_{\text{SAttn}}(F_{T1}). \quad (8)$$

Next, the refined multiscale visual features F_{vt2} are fed into the core LAE operation for candidate region selection. Specifically, a meshgrid operation is first applied to each spatial location of F_{vt2} to generate reference coordinate vectors. This operation constructs a dense reference grid over the feature map, assigning each spatial location normalized 2-D coordinates (x, y). These base positions are then expanded by adding the corresponding width and height (w, h) of the reference boxes, yielding a tensor in the bounding-box format (x, y, w, h) with shape $\left[B,\ \sum_{i\in\{64,32,16,8\}} H_i \cdot W_i,\ 4\right]$. Then, F_{vt2} and textual features F_{T2} are fused through multimodal attention and a linear projection to obtain intermediate features F_{vt3}, which are used to predict offset values for region proposals via a multilayer perceptron (MLP). The final candidate region coordinates $F_{vq} \in \mathbb{R}^{B\times K\times 4}$ are computed by summing the predicted offsets with the base positions and selecting the top-K locations

$$F_{vt3} = \mathcal{F}_{\text{Linear}}(\mathcal{F}_{\text{MAttn}}(F_{vt2}, F_{T2})) \quad (9)$$

$$F_{vq} = \mathcal{F}_{\text{TopK}}\left(\mathcal{F}_{\text{Meshgrid}}(F_{vt2}) + \mathcal{F}_{\text{MLP}}(F_{vt3})\right). \quad (10)$$

Simultaneously, the corresponding textual class tokens $F_{Tq} \in \mathbb{R}^{B\times K\times d}$ are generated by computing the interaction between F_{vt3} and F_{T2} via matrix multiplication, followed by a top-K selection

$$F_{Tq} = \mathcal{F}_{\text{TopK}}(\mathcal{F}_{\text{Matmul}}(F_{T2}, F_{vt3})). \quad (11)$$

Through this process, candidate regions are efficiently extracted from the semantically enriched visual feature map,

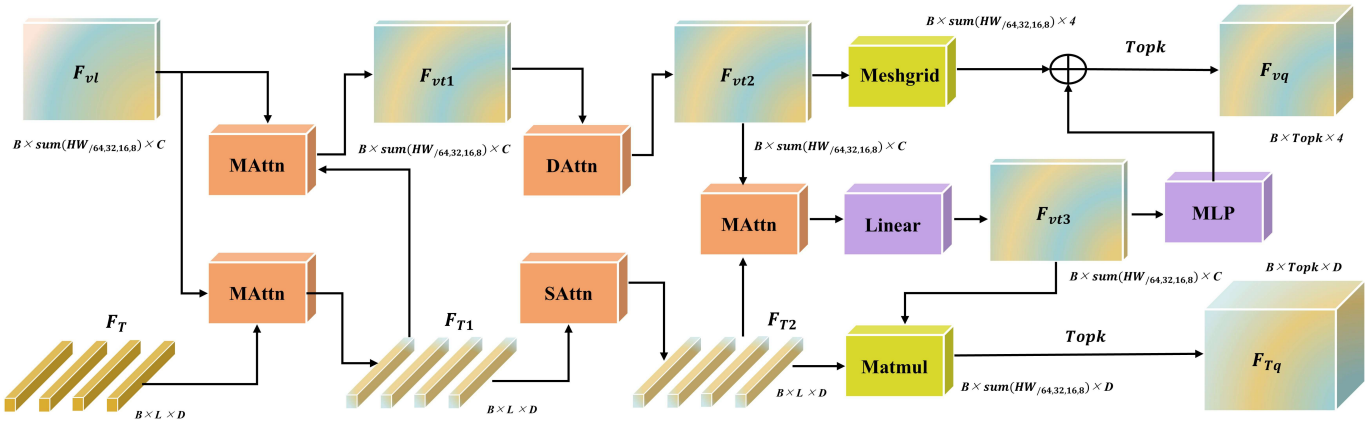


Fig. 4. Illustration of the proposed LAE module. Visual and language features serve as mutual queries for attention computation, followed by feature updating. A meshgrid operation is applied to the visual features to filter local region-specific visual representations.

guided by the textual input. The total number of candidate proposals is constrained to $K = 900$ to ensure computational efficiency. The resulting features F_{vt2}, F_{T2}, F_{vq}, and F_{Tq} are then forwarded to a multimodal Transformer decoder for subsequent reasoning and final grounding prediction.

D. Multimodal Transformer Decoder

The decoder module serves as the final processing stage of GLAED, primarily focusing on the deep interaction of local region candidate queries generated by the LAE module. Similar to the architecture of a standard Transformer decoder block, each unit in the multimodal Transformer decoder consists of a sequence of attention computations. Specifically, it first applies self-attention to the candidate queries, followed by cross attention where the queries attend to visual features, and subsequently to textual features. These attention outputs are then processed by a feed-forward network to produce updated candidate queries. The decoder consists of six such sequential units, enabling progressively deeper fusion of multimodal information. After multiple rounds of interaction, the refined local region candidate queries are passed into a three-layer MLP to perform coordinate regression. In parallel, category prediction is achieved by computing the similarity between the candidate queries and the textual features. During training, both the predicted coordinates and category are supervised using ground-truth labels, and the loss is backpropagated to guide the optimization of region selection. In this way, the decoder completes the end-to-end processing pipeline of GLAED.

IV. Dataset Construction

In this section, we first discuss the limitations of the widely used DIOR-RSVG benchmark in Section IV-A. Section IV-B describes the measures taken to construct a new RSVG dataset. Section IV-C analyzes the characteristics of the proposed DGF-RSVG dataset.

A. Problem Description

RSVG tasks are commonly evaluated using datasets such as DIOR-RSVG, which combine RS images with textual



Fig. 5. Visual comparison of image–text pairs from DIOR-RSVG and DGF-RSVG.

descriptions. However, through practical application, we have identified several limitations in this dataset.

1) Limited Semantic Richness in Textual Descriptions: As illustrated in the word cloud shown in Fig. 6(f), the descriptions primarily cover attributes such as category, color, size,

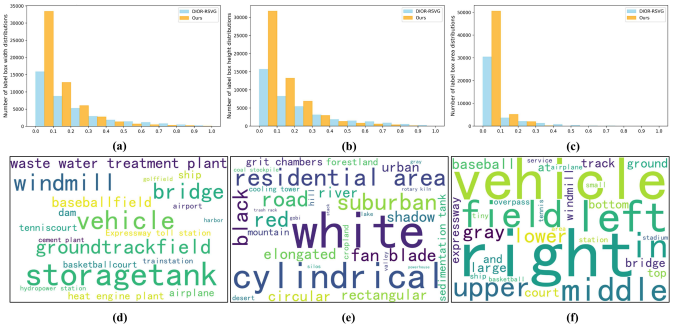


Fig. 6. Statistical comparisons between DIOR-RSVG and DGF-RSVG. Distributions of (a) bounding-box width, (b) bounding-box height, and (c) bounding-box area. Word clouds of (d) object categories and (e) object attributes in DGF-RSVG. (f) Object description words in DIOR-RSVG.

and geometric shape. However, they lack contextual awareness of the surrounding background. In particular, location descriptions are largely composed of relative position terms like left, right, and middle, without considering the spatial distribution characteristics specific to RS scenes. Statistical analysis shows that the average length of the target descriptions is only 7.47 words, suggesting limited semantic expressiveness.

2) Extremely Unbalanced Object-to-Image Area Ratios: Statistical analysis reveals that 9% of targets occupy more than 20% of the image area, while 57% occupy less than 3%. The image spatial resolution used by DIOR-RSVG is relatively high. Even if the image size can reach 800×800 pixels, some of the selected targets have a label area that is too large or too small. If it is too large, it is not conducive to expressing the background information of the target, and if it is too small, it is easy to dilute the target information.

3) Visually Simple or Ambiguous Targets: Although DIOR-RSVG includes a broad set of 20 object categories, many of these consist of isolated and visually simple instances such as storage tanks or vehicles. These targets offer limited structural and contextual diversity in high-resolution imagery. In addition, certain categories, like highway service areas, are visually similar to roadside buildings, which may lead to confusion. Meanwhile, several other common and visually distinctive types of RS targets are not sufficiently represented.

B. Optimization Measures

To address the above issues, we propose a new benchmark dataset, DGF-RSVG, built on DIOR-RSVG and enhance using high-resolution Gaofen satellite imagery. Our improvements are as follows.

1) Expand Categories With Richer Visual Features: Based on Gaofen satellite imagery, we introduce several new categories with distinctive and complex visual characteristics, including thermal power plants, cement plants, wastewater treatment plants, and hydropower stations. In DIOR-RSVG, thermal power plants and hydropower stations are oversimplified as chimneys and dams, which fails to capture the structural integrity of the former and overlooks the visual differences between dams and hydropower facilities. Wastewater treatment plants and cement plants are widely distributed and exhibit diverse internal structures while maintaining high intraclass similarity, making them valuable additions for enhancing category coverage and feature diversity. We add 1665 samples of thermal power plants, 744 of cement plants, 4000 of wastewater treatment plants, and 682 of hydropower stations. In addition to these new categories, we also extend several existing ones using Gaofen satellite imagery. Specifically, we augment storage tanks, dams, ground track fields, bridges, and wind turbines by adding 1000, 1023, 2000, 1500, and 1000 images, respectively. As shown in the DGF-RSVG part of Fig. 5, compared with the DIOR-RSVG part, the feature richness of the target and distribution scenes is significantly improved.

2) Enhanced Textual Descriptions With Geographic Context: Most textual annotations in DIOR-RSVG are relatively simple, primarily focusing on target positions within the image (e.g., middle, bottom, left, and right) or relative locations (e.g., above and below). Such expressions provide limited semantic value, especially across diverse scenes and spatial resolutions. Moreover, the descriptions largely ignore geographic contextual information, which is essential for interpreting RS imagery. In practice, the spatial distribution of targets in RS is closely linked to geographic environments. For instance, ships are typically found in water bodies or ports, bridges span rivers or valleys, and ground track fields are often located in urban residential areas. Sewage treatment plants tend to be near industrial zones and rivers for efficient discharge, while cement plants are commonly situated near mountains for raw material access.

To improve the relevance and informativeness of the descriptions, we adopted a structured annotation strategy: 1) analyze the object's inherent attributes, such as category, size, color, shape, or composition; 2) identify the geographic context or surrounding elements, such as farmland, forest, waterbody, river, mountain, or built-up area; and 3) compose a single sentence that describes the object by combining its intrinsic properties and contextual background (e.g., "A sewage treatment plant surrounded by farmland and close to a river"). As shown in Fig. 5, the image–text pairs in DGF-RSVG contain significantly more context-aware descriptions compared with DIOR-RSVG. Fig. 6(d) and (e) further presents word cloud visualizations of target category terms and attribute descriptions, respectively. On average, each target description in DGF-RSVG contains 10.7 words, compared with 7.47 words in DIOR-RSVG, reflecting a significant increase in semantic richness.

3) Remove Ambiguous or Unbalanced Categories: To improve annotation clarity and label consistency, we removed several categories that are either ambiguous or visually indistinct. Specifically, highway service areas resemble ordinary roadside buildings, stadiums often appear indistinguishable from ground track fields, overpasses are difficult to separate from bridges, and chimneys lack structural integrity as standalone targets. In addition, we excluded samples where the target occupies an extremely large or small portion of the image, which can hinder effective context learning. As a result, 14496 images were retained from the original 17402. As illustrated in Fig. 6(a)–(c), DGF-RSVG contains significantly more targets with moderate width, height, and area

ratios, while reducing the proportion of extreme cases. This refinement improves the overall distribution balance of target sizes within the dataset.

C. Dataset Implementation

Based on the improvements above, we construct the DGF-RSVG dataset by expanding upon DIOR-RSVG and integrating Gaofen satellite imagery. The dataset introduces four new categories, thermal power plants, cement plants, wastewater treatment plants, and hydropower stations and augments five existing categories while removing four ambiguous ones. In total, 12420 new images are added, resulting in a dataset with 26916 images, each with a resolution of 800×800 pixels. DGF-RSVG contains 20 object categories, 57766 annotated bounding boxes, and 62597 image–text pairs, with an average description length of 10.7 words per target.

We visualize the distribution of object width–height ratios, area proportions, and category frequencies. As shown in Fig. 6, our dataset includes more objects with a width–height ratio between 0.3 and 0.6, and the high proportion of samples with a width–height ratio below 0.1 is due to the inclusion of densely packed storage tanks in port areas. Textual analysis reveals that DGF-RSVG provides richer structural and geographic context compared with DIOR-RSVG. The dataset is randomly split into training, validation, and testing sets in a 4:1:5 ratio, following the data partition strategy originally adopted in the DIOR-RSVG dataset. In this setting, the testing set is slightly larger than the training set to ensure sufficient samples for reliable generalization evaluation.

V. Experiments

In this section, we conduct comprehensive experiments to validate the effectiveness of the proposed GLAED. Section V-A introduces the datasets and evaluation metrics. Section V-B details the implementation settings. Section V-C presents comparisons with state-of-the-art methods on DIOR-RSVG and DGF-RSVG. Section V-D reports ablation studies on key modules.

A. Datasets and Evaluation Metrics

We evaluate the GLAED detector on both the existing DIOR-RSVG and our proposed DGF-RSVG datasets. To ensure consistency and fairness, both datasets are split into training, validation, and testing sets in a 4:1:5 ratio, as in the original DIOR-RSVG setting, where the larger proportion of the testing set facilitates a more reliable assessment of model generalization. We adopt the evaluation metrics from MGVLF to quantitatively assess the model performance on both the DIOR-RSVG and DGF-RSVG datasets. The metrics include intersection over union (IoU) and Precision@T. IoU measures the overlap between the predicted bounding box and the ground-truth box. Precision@T is calculated under different IoU thresholds $T \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$, resulting in evaluation metrics, such as Precision@0.5, Precision@0.6, Precision@0.7, Precision@0.8, and Precision@0.9. In addition, we report both mean IoU and the cumulative IoU (cumIoU), based on different aggregation strategies of the IoU scores. The definitions are given as follows:

$$\text{meanIoU} = \frac{1}{N}\sum_{n=1}^{N}\frac{I_n}{U_n} \tag{12}$$

$$\text{cumIoU} = \frac{\sum_{n=1}^{N} I_n}{\sum_{n=1}^{N} U_n} \tag{13}$$

where I_n and U_n denote the intersection and union between the predicted bounding box and the ground-truth box for the nth sample and N is the total number of samples. The mean IoU is the average IoU over all samples, while the cumulative IoU is the ratio of the total intersection area to the total union area across the entire dataset.

B. Implementation Details

GLAED uses a ResNet-50 backbone pretrained on ImageNet [48] to initialize the visual feature extractor for RS imagery. All input images are resized to 640×640 pixels. For text encoding, we use the "BERT-base-uncased" model released by Google team. The input descriptions are first tokenized into word embeddings, with a maximum sequence length of 40, and tokens beyond this limit are truncated. The pretrained BERT encoder generates text features of dimension 768, which are then linearly projected to a unified 256-D space. The number of GME modules for cross-modal feature interaction is set to 4. The LAE module generates 900 query regions as input to the decoder. The multimodal decoder consists of six stacked attention layers.

GLAED is trained in a distributed setting on a server running Ubuntu 18.04, equipped with four NVIDIA RTX 4090 GPUs, each with 24 GB of memory, and PyTorch-based libraries. The model is trained for 120 epochs with a batch size of 32 on both the DIOR-RSVG and DGF-RSVG datasets. The AdamW optimizer [49] is used with a weight decay of 1×10^{-4} and a learning rate of 1×10^{-4}. The overall loss function comprises classification and localization components. The classification loss $\mathcal{L}_{\text{cls}}$ is implemented using focal loss [50]. The localization loss consists of Smooth $L1$ loss ($\mathcal{L}_{L1}$) [51] and GIoU loss ($\mathcal{L}_{\text{GIoU}}$) [52]. The total loss is defined as follows:

$$\mathcal{L} = r_c \cdot \mathcal{L}_{\text{cls}}(c, p_i) + r_l \cdot \mathcal{L}_{L1}(b, b_i) + r_g \cdot \mathcal{L}_{\text{GIoU}}(b, b_i) \tag{14}$$

where c denotes the input text query, b is the ground-truth bounding box, b_i is the predicted bounding box, and p_i is the matching probability between the text and the predicted region. The weights are set as $r_c = 2$, $r_l = 5$, and $r_g = 2$, respectively.

C. Comparison With Sota Methods

To demonstrate the effectiveness of the proposed GLAED, we conduct comparative experiments with several state-of-the-art RSVG methods, including MGVLF, LQVG, LPVA, MSVG, and Efficient Grounding DINO, on the testing sets of both DIOR-RSVG and DGF-RSVG.

TABLE I

COMPARATIVE EXPERIMENTS ON THE STATE-OF-THE-ART METHODS ON THE DIOR-RSVG TESTING SET. "PARAMS" DENOTES THE NUMBER OF MODEL PARAMETERS (IN MILLIONS). THE BEST SCORES ARE SHOWN IN BOLD

Method	Venue	Visual Enc.	Text Enc.	Params(M)	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	meanIoU	cumIoU
ReSC [34]	ECCV20	DarkNet-53	BERT	179.9	72.71	68.92	63.01	53.70	33.37	64.24	68.10
LBYL-Net [53]	CVPR21	DarkNet-53	BERT	163.8	73.78	69.22	65.36	50.45	19.52	65.23	76.37
TransVG [35]	ICCV21	ResNet-50	BERT	149.7	72.41	67.38	60.05	49.10	16.30	60.95	70.17
VLTVG [37]	CVPR22	ResNet-50	BERT	152.2	69.41	65.16	58.44	50.49	18.94	59.71	71.97
VLTVG [37]	CVPR22	ResNet-101	BERT	178.4	75.79	72.12	66.33	48.32	21.22	66.32	77.85
QRNet [38]	CVPR22	Swin-S	BERT	150.5	74.38	69.97	63.47	42.82	20.83	62.73	71.69
MGVLF [14]	TGRS23	ResNet-50	BERT	152.5	76.78	72.68	66.74	56.42	35.07	68.04	78.41
Grounding DINO [54]	arXiv23	ResNet-50	BERT	172.3	80.07	77.87	72.77	62.81	40.37	71.21	80.49
MSAM [39]	GRSL24	ResNet-50	BERT	153.3	73.46	68.42	62.74	41.17	23.33	63.45	71.76
LQVG [40]	TGRS24	ResNet-50	BERT	156.8	83.41	81.03	**75.91**	55.43	33.74	**74.02**	82.22
LPVA [41]	TGRS24	ResNet-50	BERT	156.2	82.27	77.44	72.25	60.98	39.55	72.23	**85.11**
QAMFN [42]	TGRS24	ResNet-50	BERT	128.4	81.67	75.15	68.56	57.43	34.51	71.78	84.55
FQRNet [55]	TGRS24	ResNet-50	BERT	137.8	77.23	73.16	67.20	56.94	35.37	68.35	78.76
Eff. Grounding DINO [44]	TGRS25	ResNet-50	BERT	169.3	83.05	80.60	75.00	**63.56**	42.27	73.41	81.06
GLAED(Ours)	–	ResNet-50	BERT	157.2	**84.44**	**81.52**	73.51	62.90	**43.36**	73.69	82.29

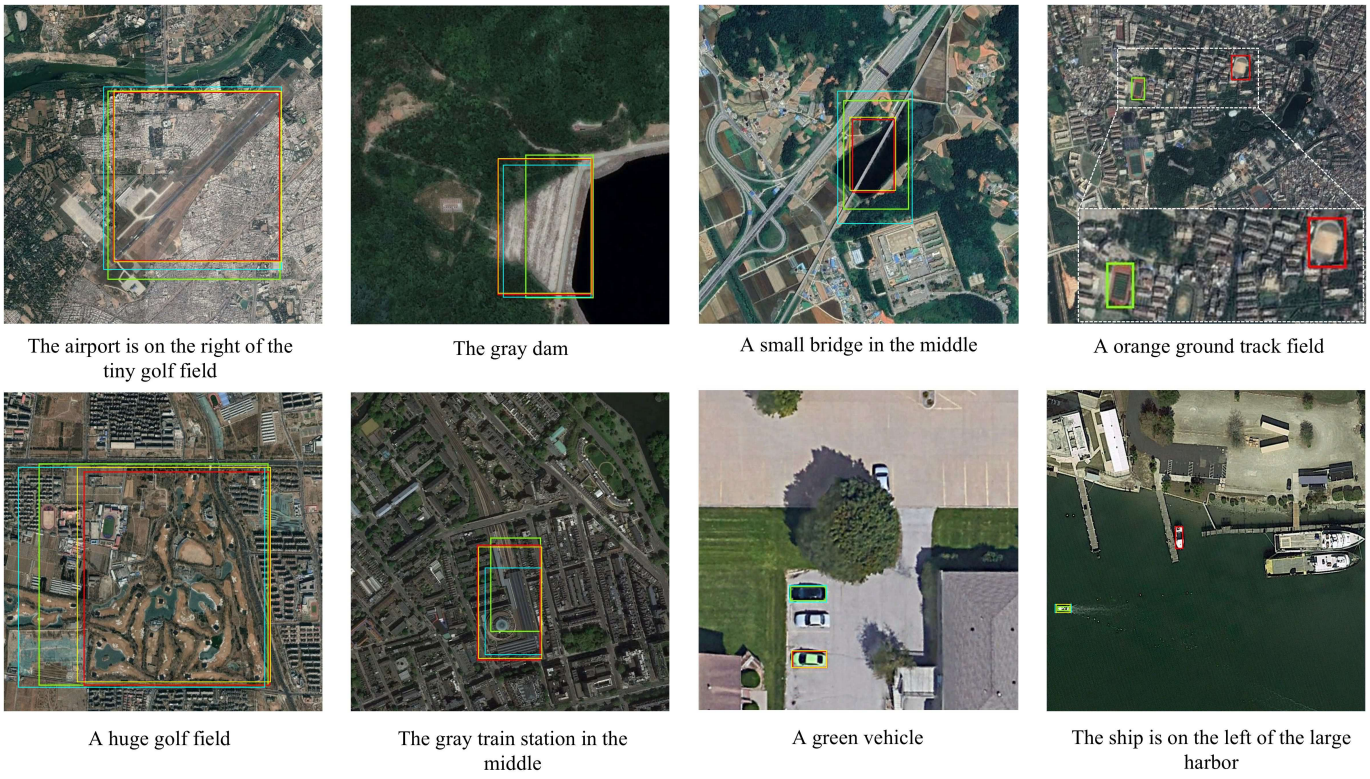


Fig. 7. Visualization comparison of our proposed GLAED with other SOTAs on DIOR-RSVG testing set. Red boxes indicate ground-truth labels, while yellow, cyan, and green boxes represent results from GLAED, LQVG, and LPVA, respectively.

1) Results on DIOR-RSVG: The quantitative results on the DIOR-RSVG testing set are presented in Table I. As shown, GLAED consistently outperforms existing methods across multiple evaluation metrics. Specifically, it achieves the highest precision of Pr@0.5, exceeding LQVG and Efficient Grounding DINO by 1.03% and 1.39%, respectively. It also achieves the best performance in Pr@0.6 and Pr@0.9, reaching 81.52% and 43.36%, respectively. In addition, the proposed model demonstrates competitive performance in terms of meanIOU and cumIOU, achieving 73.69% and 82.29%, respectively. While GLAED does not achieve the best results at Pr@0.7 and Pr@0.8, it still maintains strong performance, with only marginal differences compared with the top-performing methods. This may be attributed to the high threshold requirement, where even slight spatial deviations can affect the strict precision evaluation. Nevertheless, the consistently superior performance across a wide range of thresholds highlights the robustness and generalization ability of our method.

As shown in Fig. 7, we visualize and compare the inference results of multiple models on the DIOR-RSVG testing set, including cases of bounding-box misalignment and incorrect localization. The compared algorithms include GLAED, LQVG, and LPVA, which are represented by yellow, cyan, and green boxes, respectively, while red denotes the ground truth. From the figure, it is evident that our proposed GLAED

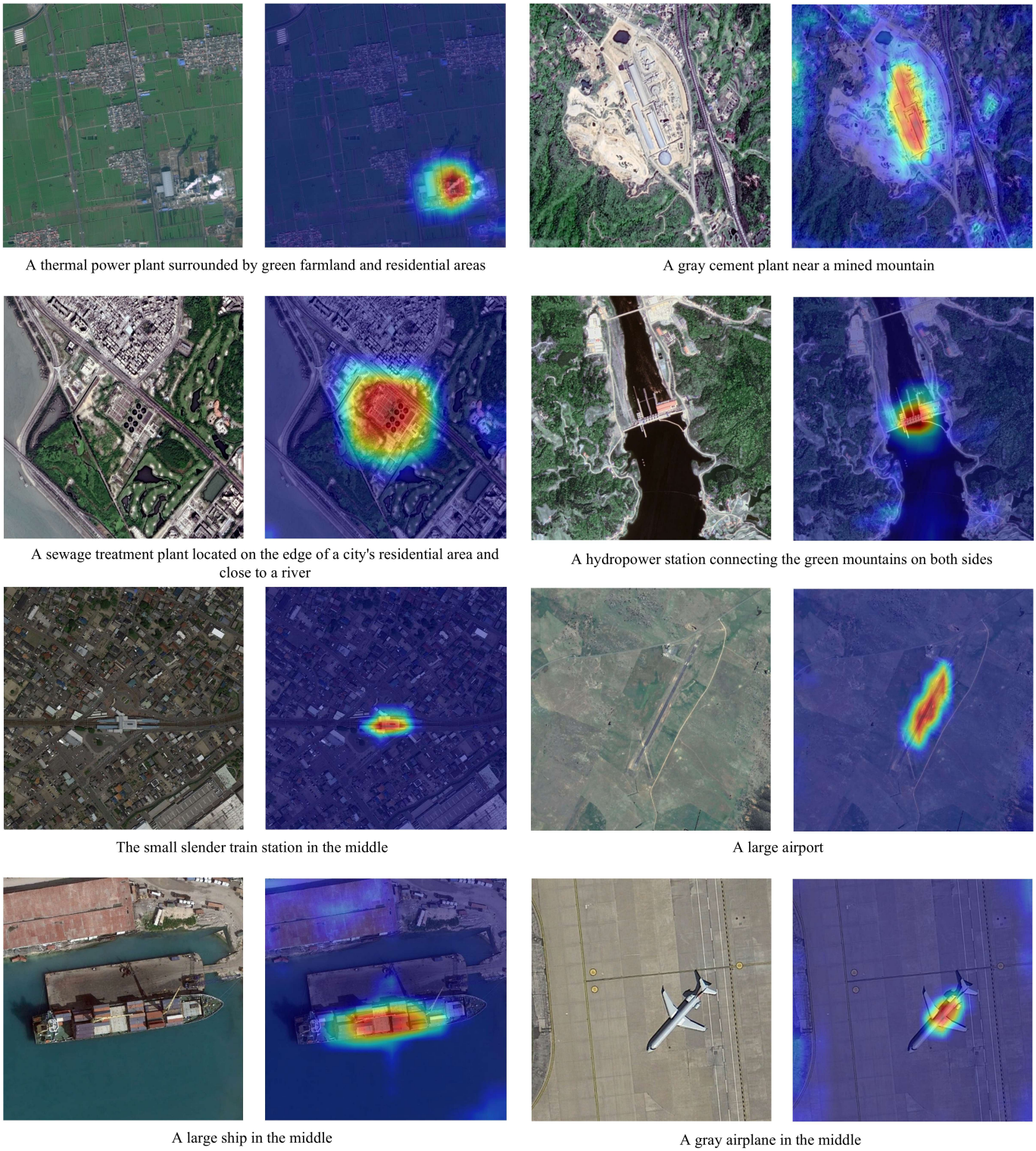


Fig. 8. Visualization of the attention maps corresponding to visual features in the inference results of the proposed GLAED on the DGF-RSVG testing set.

achieves overall superior performance. For example, in RS images corresponding to the language expressions "the airport on the right of the tiny golf field," "the gray dam," and "the gray train station in the middle," GLAED successfully learns to predict bounding boxes that more closely align with the ground truth. Moreover, in the image described as "a green vehicle," only GLAED is able to correctly localize the intended object, whereas all other methods fail. These observations demonstrate that GLAED not only achieves accurate localization but also exhibits better robustness against false positives. Nevertheless, similar to other methods, GLAED still encounters some failure cases. For example, it fails to

TABLE II

COMPARATIVE EXPERIMENTS ON THE STATE-OF-THE-ART METHODS ON THE DGF-RSVG TESTING SET. "PARAMS" DENOTES THE NUMBER OF MODEL PARAMETERS (IN MILLIONS). THE BEST SCORES ARE SHOWN IN BOLD

Method	Venue	Visual Enc.	Text Enc.	Params(M)	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	meanIoU	cumIoU
TransVG [35]	ICCV21	ResNet-50	BERT	149.7	55.11	48.54	42.65	28.01	14.60	42.44	53.46
VLTVG [37]	CVPR22	ResNet-50	BERT	152.2	56.80	52.26	45.75	30.34	16.52	45.09	56.33
QRNet [38]	CVPR22	Swin-S	BERT	150.5	58.80	53.67	47.56	39.83	19.75	44.54	55.89
MGVLF [14]	TGRS23	ResNet-50	BERT	152.5	58.37	56.93	49.67	46.36	27.04	49.03	60.12
Grounding DINO [54]	arXiv23	ResNet-50	BERT	172.3	58.75	56.65	55.80	53.66	33.46	49.79	58.76
LQVG [40]	TGRS24	ResNet-50	BERT	156.8	66.75	62.02	56.27	51.60	29.43	56.83	65.34
LPVA [41]	TGRS24	ResNet-50	BERT	156.2	61.76	58.20	53.49	48.56	26.35	51.43	63.88
GLAED(Ours)	–	ResNet-50	BERT	157.2	**71.76**	**66.14**	**59.16**	**53.47**	**35.39**	**61.13**	**70.19**

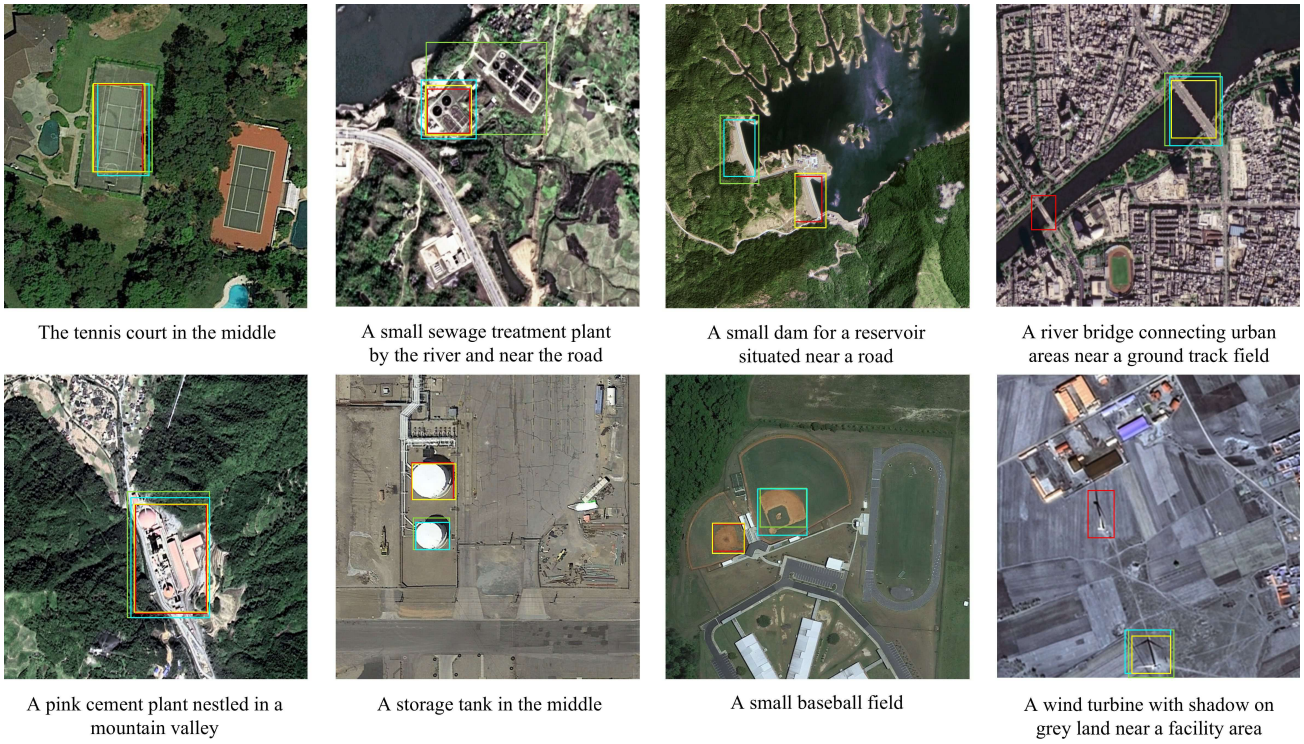


Fig. 9. Visualization comparison of our proposed GLAED with other advanced methods on DGF-RSVG testing set. Red boxes indicate ground-truth labels, while yellow, cyan, and green boxes represent results from GLAED, LQVG, and LPVA, respectively.

correctly localize the target in the images associated with the descriptions "an orange ground track field" and "the ship on the left of the large harbor."

2) Results on DGF-RSVG: We further conduct comparative experiments on the DGF-RSVG testing set, with the results summarized in Table II. We select several state-of-the-art methods with publicly available implementations, including LQVG, LPVA, Grounding DINO, TransVG, VLTVG, and MGVLF. The results show that GLAED consistently achieves superior performance across all evaluation metrics. Specifically, it surpasses the second-best method by 5.01% in Pr@0.5 and by 4.3% in mean IoU. Fig. 8 shows visualizations of the attention maps generated from visual features during the inference process. These visualizations indicate that GLAED accurately focuses on the target regions described by the text and effectively suppresses background noise. This demonstrates that GLAED successfully associates text features containing geographic information with the relevant visual features, enabling precise target localization in complex background conditions. These results further confirm the effectiveness and robustness of the proposed method.

In addition, as shown in Fig. 9, we visualize the comparison between GLAED and other advanced methods on the DGF-RSVG testing set, along with the DIOR-RSVG results in Fig. 7. When the target appears as a single object, it is generally easier to localize (e.g., Fig. 7—airport and railway station), where differences among models mainly concern positional consistency, and GLAED achieves more accurate localization. In multiobject scenes, the outcome depends largely on the visual distinctiveness of the target and the appropriateness of its textual description. When both are favorable, localization is usually accurate (e.g., Fig. 7—vehicle and Fig. 9—sewage treatment plant), and GLAED maintains a relative advantage. When the description lacks discriminative detail, predictions tend to shift toward more visually salient

TABLE III
ABLATION EXPERIMENTS OF GME AND LAE MODULES

Cases	GME	LAE	Params (M)	Pr@0.5
1	×	×	152.8	68.65
2	✓	×	155.8	70.69
3	×	✓	154.2	69.89
4	✓	✓	157.2	71.76

TABLE IV
ABLATION EXPERIMENTS WITH GME MODULES SET AT DIFFERENT LOCATIONS

Cases	Fv/4	Fv/8	Fv/16	Fv/32	Params (M)	Pr@0.5
1	✓	×	×	×	155.0	70.08
2	✓	✓	×	×	155.7	70.46
3	✓	✓	✓	×	156.5	71.32
4	✓	✓	✓	✓	157.2	71.76

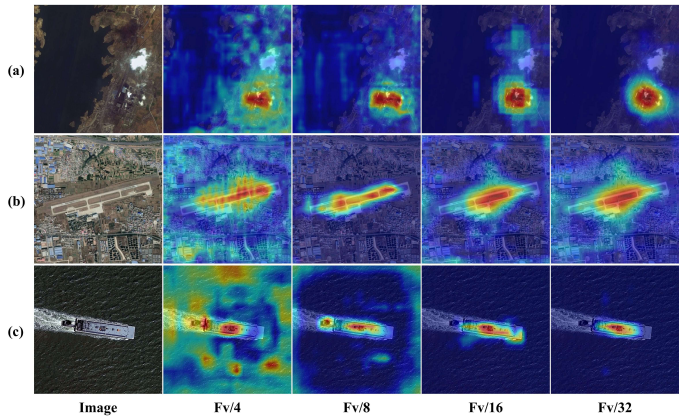


Fig. 10. Visualization of the attention maps of the GME module at different locations. Text descriptions (a) "the thermal power plant emitting smoke near the lake," (b) "the orange and gray airport in the middle," and (c) "the large ship."

distractors, causing performance fluctuations, especially within the same category (e.g., Fig. 9—bridge and storage tank). Low-visibility targets, due to small size or blurred features, are challenging for all models, even with an appropriate description (e.g., Fig. 7—ship and Fig. 9—baseball field and wind turbine). This may be because the description, though appropriate, fails to sufficiently highlight distinctive attributes. In contrast, when the description is both appropriate and specific, the success rate increases (e.g., Fig. 9—tennis court and dam), where GLAED shows clear advantages. We believe that in multiobject scenes, GLAED's GME module leverages appropriate and specific descriptions to extract high-quality visual features, enabling accurate localization even when visual salience is limited. Vague descriptions, however, have a weaker influence in GME, making LAE-generated candidates more prone to interference from salient distractors. This challenge highlights the importance of fine-grained textual annotation and model-level disambiguation for RSVG tasks.

D. Ablation Study

In this part, we conduct a series of ablation experiments on the DGF-RSVG testing set to quantitatively assess the contributions of different components in GLAED.

1) GME and LAE Modules: Table III shows the experimental results of the two core modules GME and LAE designed in the GLAED architecture. When both modules are excluded from the model, which corresponds to case 1, the Pr@0.5 drops to 68.65%, representing a decrease of 3.11% compared with the complete model in case 4, which achieves 71.76%. When only one module is included, either GME in case 2 or LAE in case 3, the GME module yields the most substantial performance gain, improving Pr@0.5 by 2.04%. The LAE module contributes a smaller improvement of 1.24%. These findings indicate that both GME and LAE are beneficial to the performance of GLAED, with GME having a more significant impact.

2) Different Locations of GME Modules: To further analyze the contribution of the GME module to the performance of GLAED, we conducted ablation experiments by varying the placement of GME modules within the network. The results, as summarized in Table IV, indicate that applying the GME module across all stages yields the most significant performance improvement. Specifically, case 3, in which the GME module is integrated at the 4×, 8×, and 16× downsampling stages, achieves the highest gain in performance. In addition, we performed attention map visualizations to examine the visual feature representations extracted by the GME module at different locations. As shown in Fig. 10, the incorporation of the GME module progressively enhances the model's attention to visual features as the downsampling ratio increases. This demonstrates the effectiveness of the GME module in improving feature interaction across scales.

TABLE V
ABLATION EXPERIMENTS OF DATTN AND CATTN IN GME MODULE

Cases	DAttn	CAttn	Params (M)	Pr@0.5
1	×	×	155.5	70.94
2	✓	×	156.2	71.27
3	×	✓	156.6	71.44
4	✓	✓	157.2	71.76

3) Structure of GME Modules: Beyond the core MAttn operation, the GME module incorporates DAttn and CAttn to process reshaped visual features and cross-modal features, respectively. We conducted experiments to assess the effectiveness of these two components. As shown in Table V, removing both modules results in a Pr@0.5 of only 70.94%, which is 0.82% lower than the full version. When adding only one component, CAttn provides a slightly greater gain than DAttn, improving performance by 0.17%. These results indicate that the use of GME along both spatial and channel dimensions enhances the model's capability in extracting informative visual features from RS images. In particular, DAttn enhances the spatial adaptability of the model by enabling content-aware, flexible sampling of relevant regions, which is especially valuable in RS scenes where objects may appear at varying scales, positions, and shapes. While the

individual performance gain from DAttn may seem modest, its contribution is crucial from a spatial modeling perspective. In addition, we investigated the role of the residual connection within GME. Removing the residual structure led to a 2.3% drop in accuracy and slower convergence, indicating that the residual structure plays a key role in stabilizing gradients and facilitating efficient optimization.

VI. Conclusion

In this article, we propose a novel RSVG network GLAED, which mainly includes a GME module for aligning image-level visual features with language features, and an LAE module for interacting local visual features with language features to enhance local attention. To overcome the limitations of DIOR-RSVG, we also construct DGF-RSVG, a new benchmark built on high-resolution Gaofen satellite imagery, which includes more diverse objects and scenes and provides richer text descriptions that highlight the semantic correlation between targets and background. Experimental results demonstrate that GLAED achieves competitive performance on both DIOR-RSVG and DGF-RSVG.

While GLAED leverages the inherent object-context associations of RS targets to advance RSVG research and improve the interpretability of RS object understanding, its applicability remains constrained by the single-target localization setting. Future studies may explore extending the framework to support more general detection tasks, such as multiobject or open-vocabulary localization. The DGF-RSVG dataset incorporates textual descriptions to express the object-context associations, which may provide a valuable reference for constructing more comprehensive RS information associations beyond the object level.

References

[1] G. E. Hinton, "Learning multiple layers of representation," *Trends Cognit. Sci.*, vol. 11, no. 10, pp. 428–434, Oct. 2007.

[2] R. Raina, A. Madhavan, and A. Y. Ng, "Large-scale deep unsupervised learning using graphics processors," in *Proc. 26th Annu. Int. Conf. Mach. Learn.*, Jun. 2009, pp. 873–880.

[3] S. Tuffery, "Natural language processing," in *Introduction to Artificial Intelligence*. Cham, Switzerland: Springer, 2023, pp. 111–186.

[4] J. Wu et al., "Towards open vocabulary learning: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 7, pp. 5092–5113, Jul. 2024.

[5] P. Xu, X. Zhu, and D. A. Clifton, "Multimodal learning with transformers: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 12113–12132, Oct. 2023.

[6] L. Xiao, X. Yang, X. Lan, Y. Wang, and C. Xu, "Towards visual grounding: A survey," 2024, *arXiv:2412.20206*.

[7] Z. Zhang, T. Zhao, Y. Guo, and J. Yin, "RS5M and GeoRSCLIP: A large scale vision-language dataset and a large vision-language model for remote sensing," 2023, *arXiv:2306.11300*.

[8] X. Li, C. Wen, Y. Hu, Z. Yuan, and X. X. Zhu, "Vision-language models in remote sensing: Current progress and future trends," *IEEE Geosci. Remote Sens. Mag.*, vol. 12, no. 2, pp. 32–66, Jun. 2024.

[9] N. Saini, A. Dubey, D. Das, and C. Chattopadhyay, "Advancing open-set object detection in remote sensing using multimodal large language model," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. Workshops (WACVW)*, Feb. 2025, pp. 414–421.

[10] P. Jia, C. Chen, D. Zhang, Y. Sang, and L. Zhang, "Semantic segmentation of deep learning remote sensing images based on band combination principle: Application in urban planning and land use," *Comput. Commun.*, vol. 217, pp. 97–106, Mar. 2024.

[11] R. Wang et al., "Remote sensing application in ecological restoration monitoring: A systematic review," *Remote Sens.*, vol. 16, no. 12, p. 2204, Jun. 2024.

[12] R. Ahmad, "Smart remote sensing network for disaster management: An overview," *Telecommun. Syst.*, vol. 87, no. 1, pp. 213–237, Sep. 2024.

[13] Y. Sun, S. Feng, X. Li, Y. Ye, J. Kang, and X. Huang, "Visual grounding in remote sensing images," in *Proc. 30th ACM Int. Conf. Multimedia*, 2022, pp. 404–412.

[14] Y. Zhan, Z. Xiong, and Y. Yuan, "RSVG: Exploring data and models for visual grounding on remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5604513.

[15] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[16] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 5998–6008.

[17] S. Kazemzadeh, V. Ordonez, M. Matten, and T. Berg, "ReferItGame: Referring to objects in photographs of natural scenes," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2014, pp. 787–798.

[18] V. K. Nagaraja, V. I. Morariu, and L. S. Davis, "Modeling context between objects for referring expression understanding," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springe, 2016, pp. 792–807.

[19] L. Yu, P. Poirson, S. Yang, A. C. Berg, and T. L. Berg, "Modeling context in referring expressions," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 69–85.

[20] T. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2014, pp. 740–755.

[21] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2014, pp. 580–587.

[22] K. Greff, R. K. Srivastava, J. Koutník, B. R. Steunebrink, and J. Schmidhuber, "LSTM: A search space Odyssey," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 28, no. 10, pp. 2222–2232, Oct. 2017.

[23] R. Hu, H. Xu, M. Rohrbach, J. Feng, K. Saenko, and T. Darrell, "Natural language object retrieval," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 4555–4564.

[24] L. Yu et al., "MAttNet: Modular attention network for referring expression comprehension," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1307–1315.

[25] R. Hong, D. Liu, X. Mo, X. He, and H. Zhang, "Learning to compose and reason with language tree structures for visual grounding," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 2, pp. 684–696, Feb. 2022.

[26] D. Liu, H. Zhang, Z.-J. Zha, and F. Wu, "Learning to assemble neural module tree networks for visual grounding," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4673–4682.

[27] P. Wang, Q. Wu, J. Cao, C. Shen, L. Gao, and A. van den Hengel, "Neighbourhood watch: Referring expression comprehension via language-guided graph attention networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 1960–1968.

[28] S. Yang, G. Li, and Y. Yu, "Dynamic graph attention for referring expression comprehension," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4644–4653.

[29] X. Chen, L. Ma, J. Chen, Z. Jie, W. Liu, and J. Luo, "Real-time referring expression comprehension by single-stage grounding network," 2018, *arXiv:1812.03426*.

[30] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.

[31] M. Schuster and K. K. Paliwal, "Bidirectional recurrent neural networks," *IEEE Trans. Signal Process.*, vol. 45, no. 11, pp. 2673–2681, Nov. 1997.

[32] Z. Yang, B. Gong, L. Wang, W. Huang, D. Yu, and J. Luo, "A fast and accurate one-stage approach to visual grounding," in *Proc. Int. Conf. Comput. Vis.*, Oct. 2019, pp. 4683–4693.

[33] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Hum. Lang. Technol.*, Minneapolis, MI, USA, vol. 1, Jun. 2019, pp. 4171–4186.

[34] Z. Yang, T. Chen, L. Wang, and J. Luo, "Improving one-stage visual grounding by recursive sub-query construction," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 387–404.

[35] J. Deng, Z. Yang, T. Chen, W. Zhou, and H. Li, "TransVG: End-to-end visual grounding with transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 1769–1779.

[36] J. Deng et al., "TransVG++: End-to-end visual grounding with language conditioned vision transformer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 11, pp. 13636–13652, Nov. 2023.

[37] L. Yang, Y. Xu, C. Yuan, W. Liu, B. Li, and W. Hu, "Improving visual grounding with visual-linguistic verification and iterative reasoning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 9489–9498.

[38] J. Ye et al., "Shifting more attention to visual backbone: Query-modulated refinement networks for end-to-end visual grounding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 15481–15491.

[39] F. Wang, C. Wu, J. Wu, L. Wang, and C. Li, "Multistage synergistic aggregation network for remote sensing visual grounding," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[40] M. Lan, F. Rong, H. Jiao, Z. Gao, and L. Zhang, "Language query-based transformer with multiscale cross-modal alignment for visual grounding on remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5626513.

[41] K. Li, D. Wang, H. Xu, H. Zhong, and C. Wang, "Language-guided progressive attention for visual grounding in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5631413.

[42] C. Li et al., "Injecting linguistic into visual backbone: Query-aware multimodal fusion network for remote sensing visual grounding," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5637814.

[43] R. Hang, S. Xu, and Q. Liu, "A regionally indicated visual grounding network for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5647411.

[44] Z. Hu et al., "Efficient grounding DINO: Efficient cross-modality fusion and efficient label assignment for visual grounding in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5609414.

[45] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[46] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*.

[47] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[48] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.

[49] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2017, *arXiv:1711.05101*.

[50] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[51] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[52] H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. Reid, and S. Savarese, "Generalized intersection over union: A metric and a loss for bounding box regression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 658–666.

[53] B. Huang, D. Lian, W. Luo, and S. Gao, "Look before you leap: Learning landmark features for one-stage visual grounding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 16883–16892.

[54] S. Liu et al., "Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2024, pp. 38–55.

[55] E. Zhao, Z. Wan, Z. Zhang, J. Nie, X. Liang, and L. Huang, "A spatial-frequency fusion strategy based on linguistic query refinement for RSVG," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5409413.